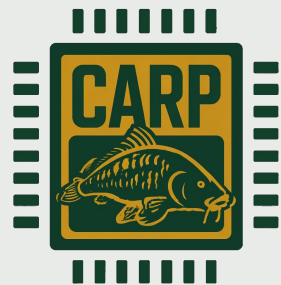
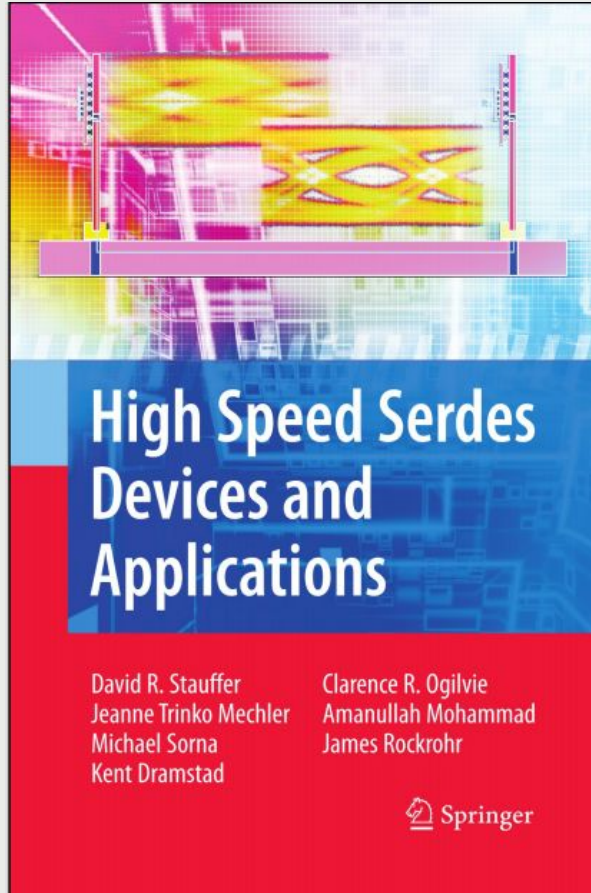


Intro to High Speed SerDes

Rudy Magana





ECEN 720: High-Speed Links Circuits and Systems

Sam Palermo
Texas A&M University

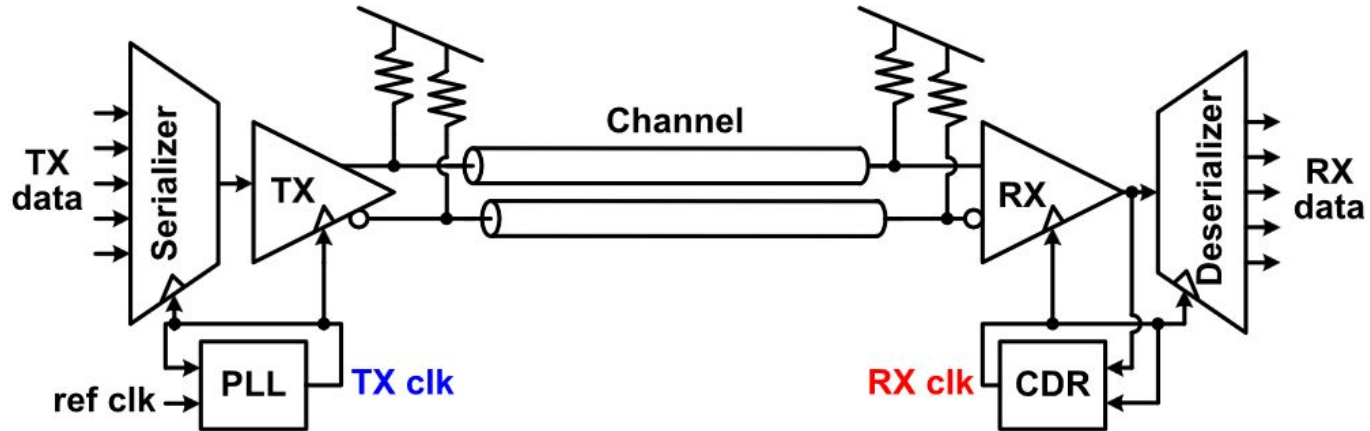
<https://people.engr.tamu.edu/spalermo/ecen720.html>

- The Book is free to download for free on the Cal Poly Library website

First, what is SerDes?

- SerDes stands for Serialize and Deserialize
- Good for Chip to Chip communication
- Fast. Less Wires.

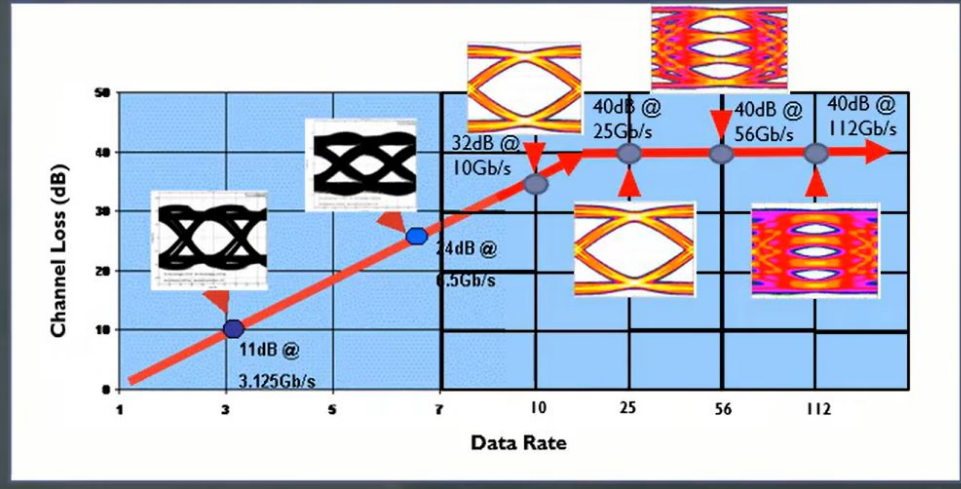
8 bit example



Interface Speeds over Last 20 Years



Channel Losses/Modulation Schemes Versus Data Rate



Why are we making them faster? What's sparking the boom? (AI Data Centers.)

<https://www.youtube.com/watch?v=OY2Dn4EDPiA>

Why are we using SerDes?

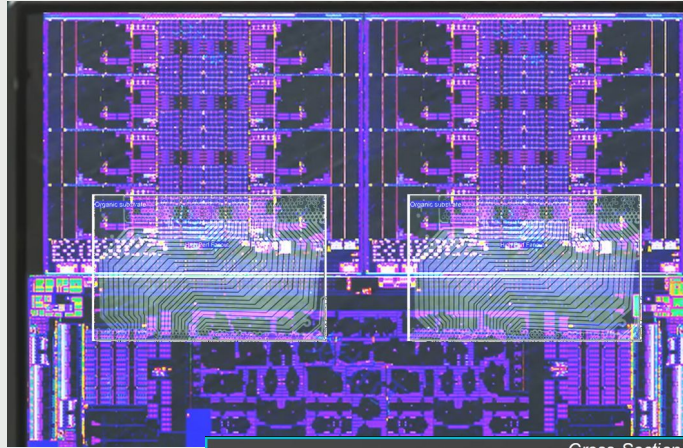
Too many wires.



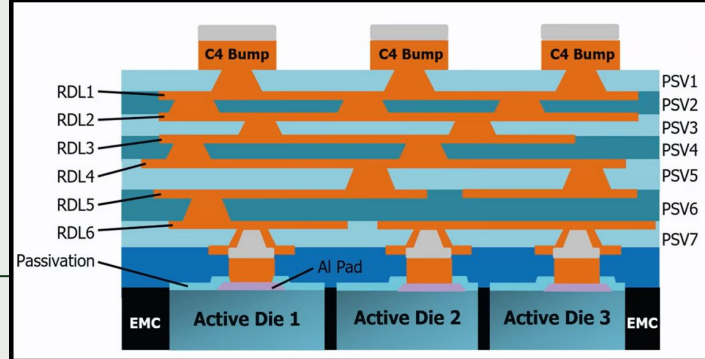
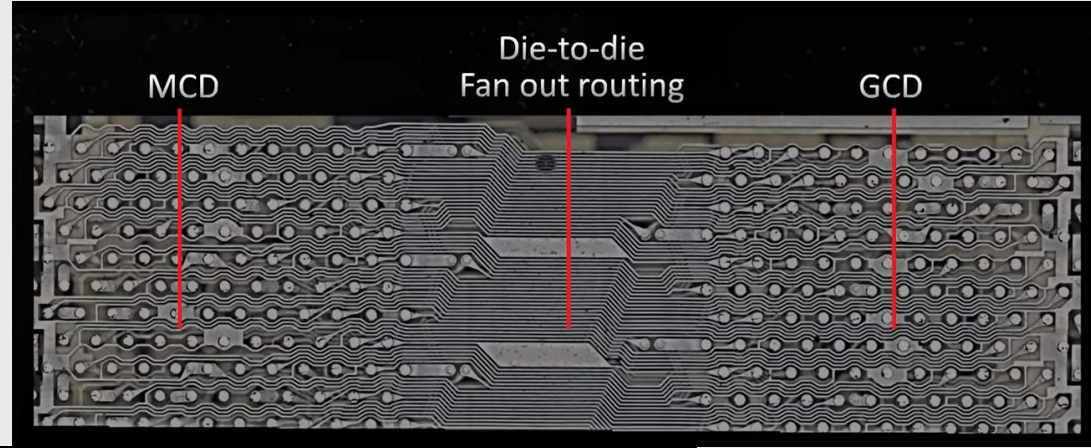
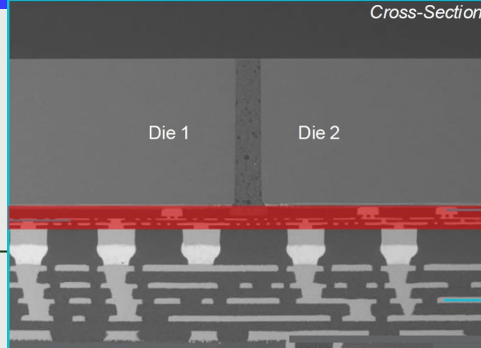
Just some interesting and cool pictures

AMD Strix Halo Chip uses state of the art Fanout Technology (TSMC)

<https://www.youtube.com/watch?v=maH6KZ0YkXU&t=137s>

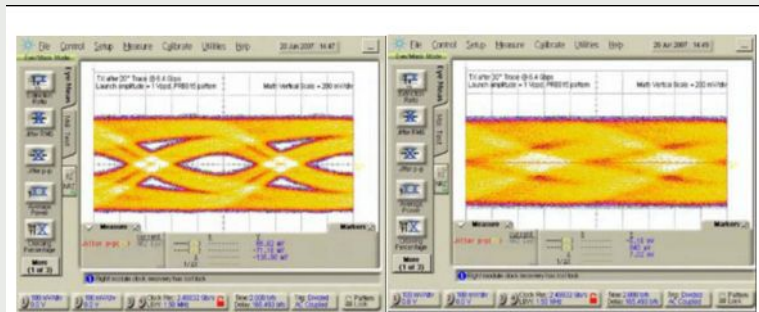


Cross-Section



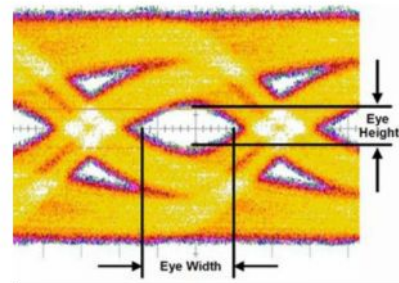
Topic 1. Eye Diagrams

- We want a fast clock, and
 - We want to tell between 0 and 1 reliably, and
 - We want a realistic and reliable time to sample
-
- We overlay the transitions from HIGH to LOW on top of each other
-
- But how to do we find to most optimal sample time?



(a) Open Data Eye

(b) Closed Eye

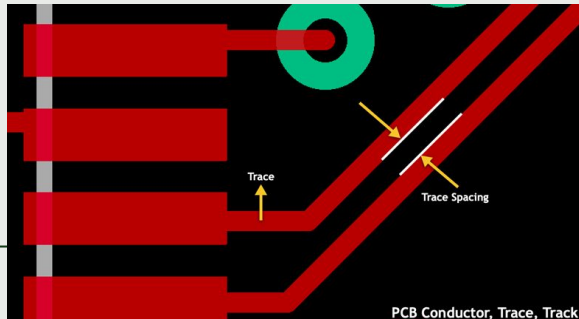


(c) Measuring the Eye Opening

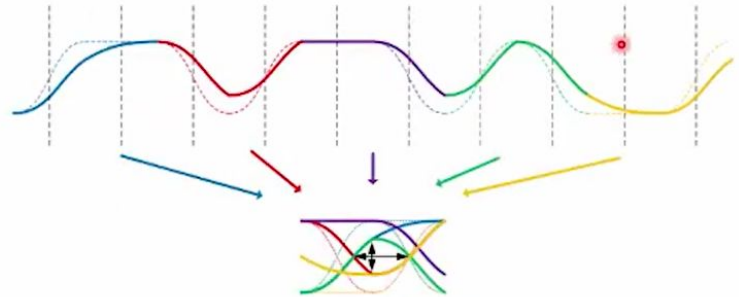
Issues: Inter Signal Interference (ISI) Electromagnetic Interference (EMI)

- We call this topic **Signal Integrity**
- Depending on the main Transistor Type of your Chip. Your circuit may pull up to Vdd better, or Drain to GND better
 - Pmos, Nmos, Cmos?

How do we combat this?



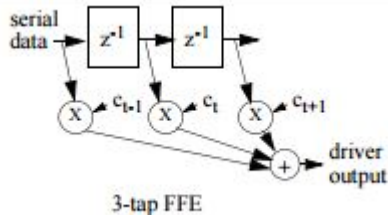
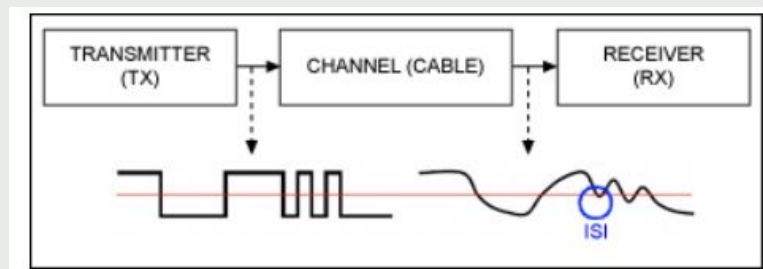
Intersymbol Interference (ISI)



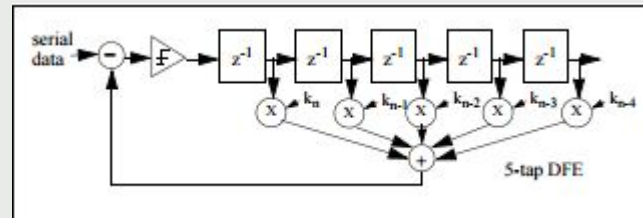
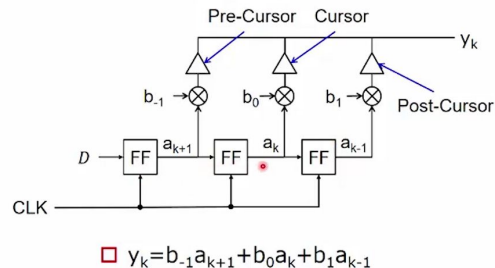
Topic 2: Equalization

FIR = Finite Impulse Response
FFE = Feed Forward Equalizer
DFE = Decision Feedback Equalizer

- Using Signal History to compensate for ISI (Inter Signal Interference)



TX FIR Equalization

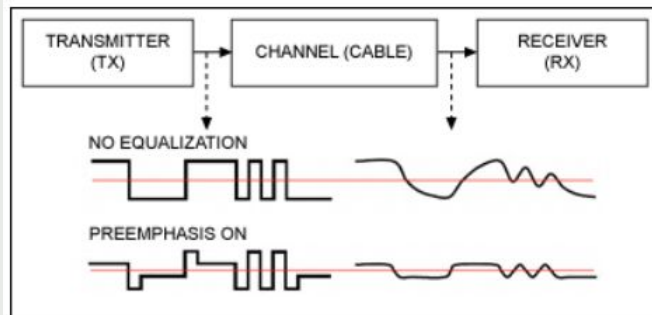


Pre-Emphasis (Tx)

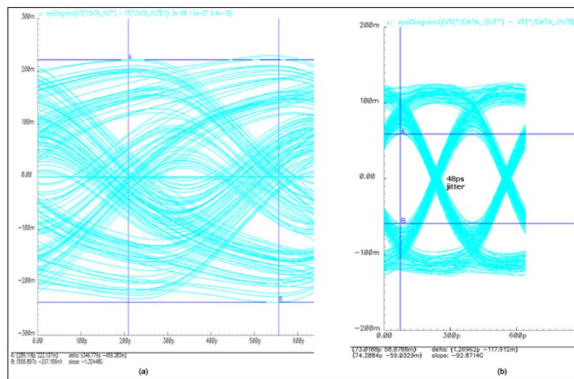
and

Post-Emphasis (Rx)

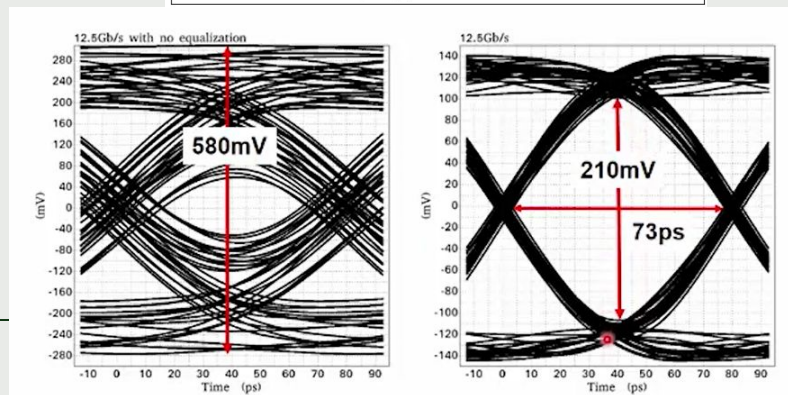
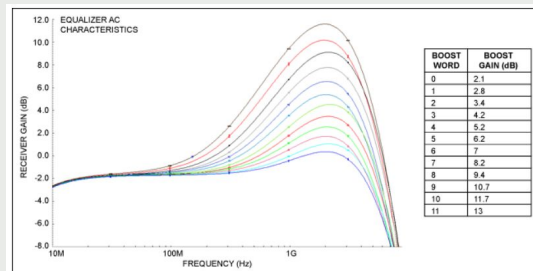
- Modify the Waveform
 - $\text{Tx_Eq} + \text{Waveform} = \text{Desired Signal}$



preemphasis (Figure 4).



- Use previous detected bits to boost or subtract ISI
 - Boost (Amplify) on long wires (Signal Dying)



Topic 3: Clocks

Source Synchronous Clocking

vs

Clock Recovery

- Clock Sent by Tx (w/ data)
- Jitter
- More ISI, EMF... Why?

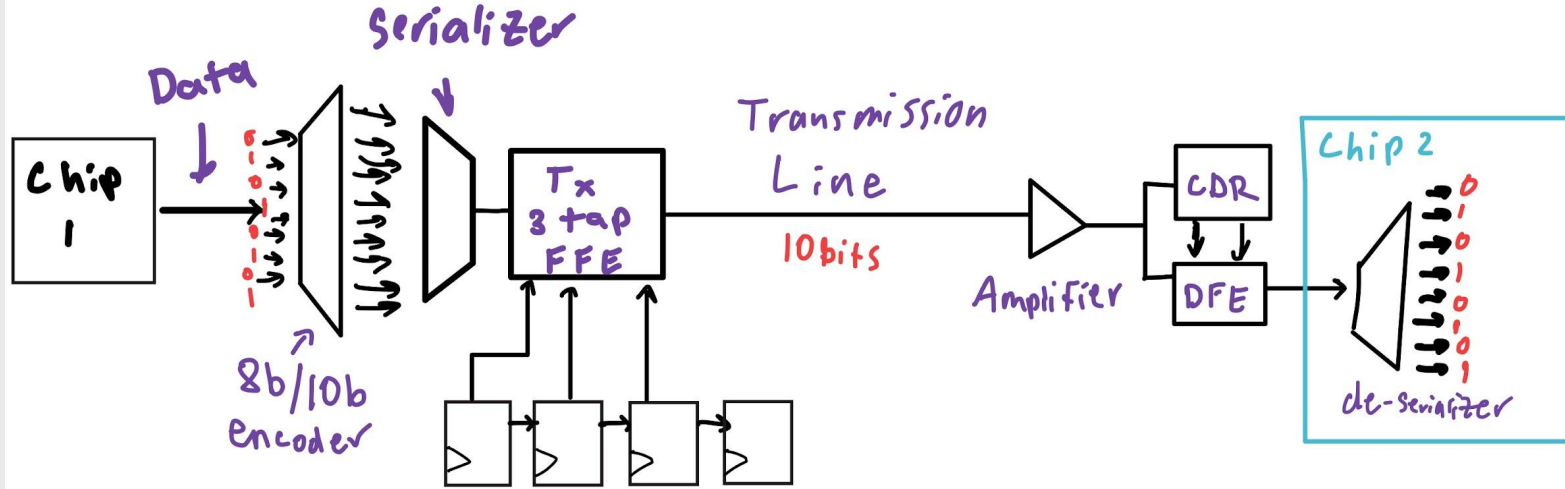
- Clock Recovered/Created in Rx
 - Data Transition Edges give CLK Freq Info
 - A whole topic in and of itself (that I still don't know)

Clock Jitter

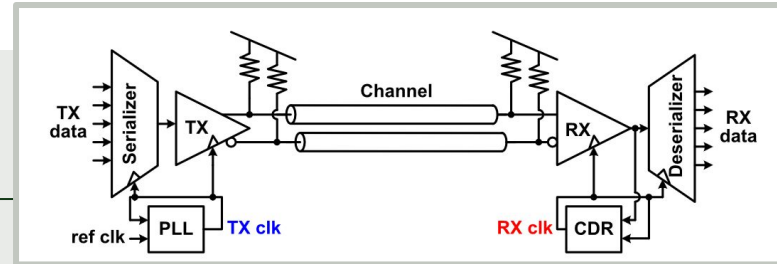
Phase Lock Loops (PLLs)

Startup Sequences -> Encoding Schemes

Design and Architect

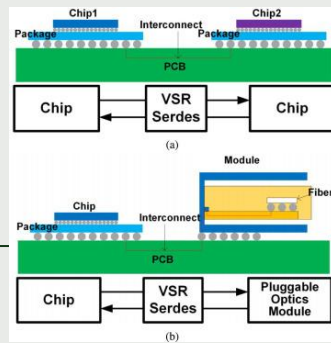
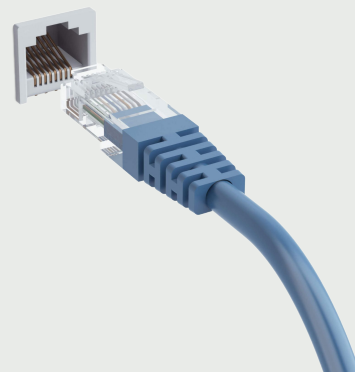
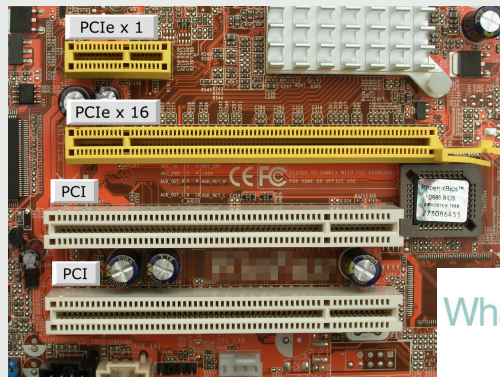


!! Add wires from registers to DFE, I forgot. Also De-Serializer, whoops.



Where is this used?

- I2C
- Ethernet
- PCIe
- UCIe
- GMSL (Automotive)
- UART



Congrats!!! You
are about here.



Table of Contents

Chapter 1 - SerDes Concepts

Chapter 2 - HSS Features and Functions

Chapter 3 - HSS Architecture and Design

Chapter 4 - Protocol Logic and Specifications

Chapter 5 - Overview of Protocol Standards

Chapter 6 - Reference Clocks

Chapter 7 - Test and Diagnostics

Chapter 8 - Signal Integrity

Chapter 9 - Power Analysis

Chapter 10 - Chip Integration

Thank You!